CENG381 Stochastic Processes

Irreducibility and Periodicity - Question Set 1

Q1 (30 points). Consider the Markov chain with transition matrix:

$$\begin{array}{cc} & \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{array}{c} 0 \\ 1 \\ 2 \\ 3 \end{array} & \begin{bmatrix} 0.0 & 1.0 & 0.0 & 0.0 \\ 0.5 & 0.0 & 0.5 & 0.0 \\ 0.0 & 0.5 & 0.0 & 0.5 \\ 0.0 & 0.0 & 1.0 & 0.0 \end{bmatrix} \end{array}$$

a) Draw the transition diagram for this chain.

b) Is the chain irreducible? Justify your answer by identifying whether every state can reach every other state.

c) Compute the period of each state. (Recall: the period of state x is d(x) = gcd{t >= 1 : P^t(x,x) > 0}.) Is the chain aperiodic?

Q2 (35 points). Consider the random walk on a 4-cycle: states {0, 1, 2, 3} arranged in a cycle, where from any state the chain moves clockwise or counter-clockwise each with probability 1/2.

a) Write the transition matrix P for this chain.

b) Show that P^2 has a block structure. What does this tell you about the period of the chain?

c) Now consider the lazy version: at each step, with probability 1/2 the chain stays put, and with probability 1/2 it takes a step on the 4-cycle. Write the new transition matrix Q = (1/2)I + (1/2)P and show it is aperiodic.

Q3 (35 points). The PageRank algorithm models web surfing as a Markov chain on the graph of web pages. It requires the chain to be both irreducible and aperiodic to guarantee convergence to a unique stationary distribution.

a) Consider a web graph with 4 pages and transition matrix:

$$\begin{array}{cc} & \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} & \begin{bmatrix} 0.0 & 1.0 & 0.0 & 0.0 \\ 0.5 & 0.0 & 0.5 & 0.0 \\ 0.0 & 0.0 & 0.0 & 1.0 \\ 1.0 & 0.0 & 0.0 & 0.0 \end{bmatrix} \end{array}$$

b) Is this chain irreducible? Identify all communicating classes.

c) The standard PageRank fix uses the 'damping factor' d = 0.85: replace P with P' =

(1-d)*(1/n)*J + d*P, where J is the all-ones matrix and n=4. Write P' and explain why P' is now irreducible and aperiodic.